ModelIt!

Student Activity Pack

The Invisible Plant Factory: Photosynthesis Deeper

How Plants Build Themselves from Thin Air and Sunlight

Grade 06L2

NGSS MS-LS1-6, MS-LS1-7

Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where do you think plants get the material to grow? Is it from the soil, the air, water, or sunlight?**

2. **If you put a plant in a dark closet with plenty of water and soil, what would happen and why?**

3. **Where does the oxygen you breathe come from? How is it made?**

4. **A tree can weigh thousands of pounds but grows from a tiny seed. Where does all that weight come from?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Light Intensity | CO_2 Concentration | Water Availability | Glucose Production | Oxygen Released | Plant Biomass Growth

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Sun

All inputs maximized

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Darkness

Lock Light to 0%, other inputs normal

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Drought

Lock Water to 10%, other inputs normal

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Temperature — Photosynthesis enzymes work fastest at optimal temperatures. Too cold or too hot slows or stops the process

Chlorophyll Concentration — More chlorophyll means more light can be captured. Pale or yellowing leaves photosynthesize less

Stomata Openness — Pores on leaves that open to let CO_2 in but also let water vapor out, creating a tradeoff between photosynthesis and water conservation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

A giant redwood tree weighs over 2 million pounds, but it started as a tiny seed. Where did all that mass actually come from? Use evidence from your simulation to support your answers.

1. **Why is it surprising that most of a tree's mass comes from air rather than soil?**

2. **If you increase light but not CO_2, why does photosynthesis eventually plateau?**

3. **How is photosynthesis the reverse of cellular respiration? What do they share?**

4. **Why do plants produce oxygen even though they do not need it?**

5. **How does understanding limiting factors help farmers grow more food?**

8 STEM Challenge: Design a Plant Growth Optimization Chamber

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YOUR ENGINEERING MISSION

Design a growth chamber that maximizes photosynthesis and plant biomass production by optimizing light, CO_2, and water delivery.

THE SCENARIO

A vertical farming company wants to grow food in urban buildings without natural sunlight. Your agricultural engineering team must design a growth chamber that delivers optimal light, CO_2, and water to maximize plant production efficiency.

Engineering Design Thinking

1. WHAT TYPE OF ARTIFICIAL LIGHT WILL YOU USE AND HOW MANY HOURS PER DAY WILL IT RUN?
2. HOW WILL YOU SUPPLEMENT CO_2 LEVELS ABOVE NORMAL ATMOSPHERIC CONCENTRATION?
3. WHAT WATERING SYSTEM WILL DELIVER WATER EFFICIENTLY WITHOUT WATERLOGGING THE ROOTS?
4. HOW WILL YOU MEASURE BIOMASS GROWTH OVER TIME TO EVALUATE YOUR CHAMBER'S EFFECTIVENESS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Plant Biologists and Agricultural Engineers study photosynthesis and design systems to optimize crop production, including vertical farms, greenhouses, and space agriculture. They earn $55,000-$120,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 A giant redwood tree weighs over 2 million pounds. Where did most of that mass come from?

- **A.** Minerals and nutrients absorbed from the soil through the roots
- **B.** Water absorbed from the ground
- **C.** Carbon dioxide gas from the atmosphere, converted to glucose and then to wood through photosynthesis
- **D.** Sunlight energy that was converted into solid matter

2 In the photosynthesis model, what happens when light is abundant and water is plentiful but CO2 is very low?

- **A.** Glucose production runs at maximum because two out of three inputs are high
- **B.** Glucose production is severely limited because CO2 is the limiting factor, regardless of how much light and water are available
- **C.** The plant switches to using nitrogen from the air instead of CO2
- **D.** The extra light compensates for the lack of CO2

3 Why is oxygen described as a 'byproduct' of photosynthesis rather than its main purpose?

- **A.** Plants do not actually produce oxygen
- **B.** Plants produce oxygen to help animals breathe, which is the main purpose
- **C.** Oxygen is released when water molecules are split to provide hydrogen and electrons for glucose production. The plant needs the hydrogen, not the oxygen.
- **D.** Oxygen is just leftover sunlight energy that the plant cannot use

4 How are photosynthesis and cellular respiration related?

- **A.** They are the same process happening in different organelles
- **B.** They are essentially opposite reactions: photosynthesis builds glucose from CO2 and water using light energy, while cellular respiration breaks glucose down into CO2 and water to release energy
- **C.** Photosynthesis only occurs in plants and cellular respiration only occurs in animals
- **D.** They have no relationship because they occur in different types of cells

5 **A scientist grows two identical plants in sealed containers. Plant A has normal light, CO_2, and water. Plant B has normal light and water but the CO_2 has been removed. After four weeks, what would the model predict?**

- **A.** Both plants grow equally because light provides all the energy needed
- **B.** Plant B grows slightly less because it can use water as an alternative carbon source
- **C.** Plant B barely grows because without CO_2, it cannot perform photosynthesis or produce glucose for building new tissue
- **D.** Plant B grows more because removing CO_2 forces the plant to use more efficient processes